Related Work and Data Description

STADIOchoice Streaming Churn Prediction, CAP182 Capstone SS2, Part A

Related Literature 1

Title	Customer churn prediction system: a machine learning approach
Harvard reference	Lalwani, P., Mishra, M.K., Chadha, J.S. and Sethi, P. (2022) 'Customer churn prediction system: a machine learning approach', Computing, 104(2), pp.271-294.
Link	https://doi.org/10.1007/s00607-021-00908-y
Modelling	Compared Logistic Regression, Naive Bayes, SVM, Decision Tree, Random Forest, AdaBoost, XGBoost, CatBoost, and Extra Trees. Used a gravitational search algorithm for feature selection ahead of model training. AdaBoost and XGBoost were the top performers (81.71% and 80.8% accuracy, 84% AUC).
Preprocessing techniques	Data cleaning and encoding, followed by gravitational search based feature selection, 80/20 train-test split.
Datasets used	IBM Telco Customer Churn dataset (via Kaggle), 7,043 customers, 21 columns, 26.53% churn rate. The same dataset selected for this project's Part A.
Key takeaways	Ensemble methods (AdaBoost, XGBoost) consistently beat simpler linear and single tree models on this exact dataset. Feature selection before modelling is worth doing rather than using every raw column. This paper's reported accuracy and AUC give a real external benchmark to compare our own results against, since we use the identical dataset.

Related Literature 2

Title	Explainable AI-driven customer churn prediction: a multi-model ensemble approach with SHAP-based feature analysis
Harvard reference	El Attar, A. and El-Hajj, M. (2026) 'Explainable AI-driven customer churn prediction: a multi-model ensemble approach with SHAP-based feature analysis', Frontiers in Artificial Intelligence, 9, 1748799.
Link	https://doi.org/10.3389/frai.2026.1748799
Modelling	Trained XGBoost, LightGBM, Gradient Boosting, Random Forest, AdaBoost, Logistic Regression, and a Multi-layer Perceptron, plus a soft-voting ensemble. XGBoost performed best (accuracy 0.84, F1 0.84, AUC-ROC 0.932).
Preprocessing techniques	Median imputation for missing TotalCharges values, StandardScaler on numeric features, one-hot encoding for categoricals, ordinal encoding for contract duration, SMOTE oversampling (k=5) applied only to the training split.
Datasets used	IBM Telco Customer Churn dataset (via Kaggle), 7,043 customers, 21 original columns expanded to 46 after preprocessing. Same dataset as this project's Part A choice.
Key takeaways	A directly reusable preprocessing recipe for the exact dataset used here: imputing TotalCharges, scaling, encoding, SMOTE for the class imbalance. The engineered features AvgMonthlyCharge and HasMultipleServices are concrete feature engineering ideas to adapt. SHAP based explanation of churn drivers is useful for framing the Part D recommendation report.

Related Literature 3

Title	Enhancing customer retention in telecom industry with machine learning driven churn prediction
Harvard reference	Sikri, A., Jameel, R., Idrees, S.M. and Kaur, H. (2024) 'Enhancing customer retention in telecom industry with machine learning driven churn prediction', Scientific Reports, 14, 13097.
Link	https://doi.org/10.1038/s41598-024-63750-0
Modelling	Compared Perceptron, Multi-layer Perceptron, Naive Bayes, Logistic Regression, KNN, Decision Tree, Gradient Boosting, and XGBoost. XGBoost with a 75:25 class ratio performed best (accuracy 89.6%, F1 68.8%).
Preprocessing techniques	Dropped non-predictive identifier columns, converted the target to binary, and tested three class balancing strategies (over-sampling, under-sampling, and a proposed ratio-based balancing) across five class ratios.

Datasets used	A separate telecom churn dataset from Kaggle (usage and call based features such as day, evening, and night minutes, and customer service calls), not the same dataset as this project's Part A choice, but the same problem type.
Key takeaways	Confirms XGBoost's edge over simpler models holds across different telecom churn datasets, not just ours, which strengthens the model choice made for this project. The finding that class balancing strategy materially changes results means imbalance handling should be tested, not applied blindly.

Publicly Available Dataset (used in Part B)

Name	Telco Customer Churn
Harvard reference	IBM (2018) Telco Customer Churn, Kaggle. Available at: https://www.kaggle.com/datasets/blastchar/telco-customer-churn (Accessed: 28 September 2026).
Link	https://www.kaggle.com/datasets/blastchar/telco-customer-churn
Number of instances	7,043 customers in a single table. No separate train/test split is provided, so one will be created for this project.
Number of feature columns	21 columns total: 1 customer ID column (dropped before modelling), 19 predictor features covering demographics, account information, services subscribed, and billing, and 1 target column (Churn).
Number of classes	2 (Churn: Yes / No). Binary classification problem.
Target range	Not applicable, this is a classification problem, not a regression problem.
Origin	Originally released as an IBM Sample Data Set for demonstrating IBM Cognos Analytics and Watson Analytics, later uploaded to Kaggle. Not synthetically generated, based on real anonymised telecom customer records.